

ESKISEHIR HAVALIMANI, Turkey

WMO: 171240

Lat: 39.781N

Lon: 30.580E

Elev: 2579

StdP: 13.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	18.0	9.1	9.6	19.9	13.6	12.0	22.0	21.8	35.0	18.9	34.5	4.4	90	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	92.3	66.0	89.5	65.3	86.4	64.4	70.3	85.3	68.7	82.9	67.2	81.1	8.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	104.2	77.1	63.9	97.8	76.1	62.3	92.5	74.6	36.0	84.2	34.4	83.1	33.2	81.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.7	14.9	DB	6.7	97.5	6.4	3.0	2.1	99.6	-1.6	101.3	-5.2	103.0	-9.8	105.1	
			WB	6.0	72.9	6.2	3.9	1.5	75.7	-2.2	78.0	-5.7	80.2	-10.2	83.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	33.3	36.7	43.2	51.3	60.1	67.1	72.4	72.7	64.7	55.0	44.3	36.0
	DBStd	15.27	8.32	8.27	7.32	7.07	5.90	4.96	4.20	4.31	5.45	6.61	7.33	7.63
	HDD50	1858	518	375	230	66	3	0	0	0	0	29	199	438
	HDD65	4978	982	792	675	411	170	35	2	3	70	316	622	900
	CDD50	3011	0	3	20	106	317	513	696	702	442	183	26	3
	CDD65	655	0	0	0	1	19	98	232	240	61	4	0	0
	CDH74	7953	0	0	4	67	398	1247	2634	2568	922	113	0	0
	CDH80	2954	0	0	0	7	73	401	1099	1065	292	17	0	0
Wind	WSAvg	6.5	6.8	6.8	7.2	6.7	6.1	6.6	7.5	7.2	6.0	5.1	5.4	6.2
Precipitation	PrecAvg	15.4	1.7	1.2	1.6	1.6	1.8	1.4	0.5	0.4	0.6	1.2	1.2	1.9
	PrecMax	20.0	4.1	3.1	3.1	4.6	5.0	5.0	2.5	1.6	3.3	3.9	2.9	5.0
	PrecMin	11.7	0.1	0.1	0.3	0.4	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	1.1	0.8	0.7	0.9	1.1	1.1	0.6	0.4	0.7	0.8	0.7	1.0
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	63.5	71.7	80.5	86.0	91.8	97.0	95.3	91.5	82.5	69.5
MCWB			46.9	50.3	51.3	58.1	60.6	65.3	68.0	66.7	64.1	63.1	53.7	50.2
2%		DB	51.6	57.4	66.0	75.3	82.0	87.8	92.9	92.1	87.1	77.0	64.6	53.6
		MCWB	45.0	47.0	50.3	55.9	60.4	65.3	66.6	65.8	62.5	59.5	53.0	47.6
5%		DB	47.9	53.6	61.7	70.6	78.7	84.5	89.5	89.3	83.0	73.0	60.5	49.9
		MCWB	43.0	45.5	48.4	53.9	60.0	63.7	66.0	65.3	61.8	57.3	50.9	45.4
10%		DB	44.5	49.7	57.5	66.3	75.1	81.5	86.1	86.1	79.2	68.3	56.8	46.7
		MCWB	41.2	43.4	46.5	52.3	58.3	62.8	65.0	65.0	60.9	55.7	49.2	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	51.5	54.6	60.7	66.6	70.0	73.7	72.7	68.8	66.3	56.5	51.1
		MCDB	54.7	60.3	67.5	75.2	80.9	84.6	90.1	89.3	83.8	78.2	64.3	56.9
	2%	WB	45.5	48.0	51.5	57.3	63.6	67.6	70.8	70.3	66.3	61.3	53.8	48.1
		MCDB	50.7	56.2	63.2	72.0	77.6	82.9	86.4	85.1	80.7	71.3	62.2	52.9
	5%	WB	43.1	45.9	49.3	55.3	61.4	65.7	68.9	68.5	64.4	59.0	51.6	46.0
		MCDB	47.0	52.7	59.8	68.9	74.0	80.8	83.3	82.3	77.9	68.5	58.8	49.3
	10%	WB	41.0	43.5	47.2	53.1	59.5	64.0	67.2	67.1	62.6	57.1	49.6	43.7
		MCDB	44.0	48.9	56.4	64.2	71.3	77.8	81.0	80.3	75.0	66.3	55.9	46.6
Mean Daily Temperature Range	5% DB	MDBR	12.2	16.8	21.1	23.4	24.6	25.3	26.6	26.4	27.0	23.4	19.6	12.5
		MCDBR	17.5	24.2	30.3	31.8	30.9	31.0	31.8	31.1	33.8	31.0	26.0	16.0
		MCWBR	12.2	15.2	16.9	15.6	13.8	12.2	11.3	10.1	13.6	15.4	15.5	10.8
	5% WB	MCDBR	16.1	22.2	26.7	28.0	27.0	27.9	27.2	26.1	29.0	25.2	22.3	14.2
		MCWBR	11.7	14.6	15.7	15.0	13.9	12.9	12.3	10.7	13.9	14.2	14.0	10.2
Clear-Sky Solar Irradiance	taub	0.314	0.341	0.387	0.425	0.404	0.383	0.380	0.382	0.366	0.358	0.326	0.314	
	taud	2.383	2.319	2.199	2.122	2.252	2.342	2.354	2.338	2.374	2.387	2.437	2.428	
	Ebn at Noon	269	276	273	270	278	283	283	280	278	267	263	260	
	Edh at Noon	27	33	42	48	43	39	39	38	35	31	25	24	
All-Sky Solar Radiation	RadAvg	579	858	1224	1585	1908	2201	2335	2053	1593	1087	751	518	
	RadStd	64	111	106	154	154	192	90	96	118	83	89	69	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-2.23	-1.84	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page